

useful for a broader range of applications than just the calculator family."

He also said, "Dr Federico Faggin was hired around in April of 1970 and given the responsibility for chip circuit design and layout, to turn this architecture into a physical transistor layout. He developed a number of techniques to take advantage of Intel's new silicon gate metal oxide silicon (MOS) process and even found ways to improve performance using techniques that others felt impossible to do with silicon gate. He had working parts by around January of 1971."

The result was the 4004 microprocessor, a 4-bit chip containing 2300 MOS transistors, and as powerful as the ENIAC. But the sidelight is that only after delivering the chip to Busicomp, Intel realised the market potential of the chip. Intel had to renegotiate with Busicomp and regain the exclusive rights. We would have missed the latest Intel Core i7 which contains 995 million transistors.

So let us salute the pioneers! Kilby wrote in the autobiography submitted to Nobel committee, "Whether the research is applied or basic, we all 'stand upon the shoulders of giants,' as Isaac Newton said. I'm grateful to the innovative thinkers who came before me, and I admire the innovators who have followed."

Kilby said, "From 1978 to 1984, I spent much of my time as a distinguished professor of electrical engineering at Texas A&M University." And his words for the honour, "The 'distinguished' part is in the eye of the beholder, and I really didn't do much 'professing'."

Was he unhappy at his late selection to the Nobel? "It's not too late—at least I'm still alive. You have to live long enough to receive the Prize," he said. Noyce could not live long enough.

But Noyce charted an American revolution by the way he managed the

two companies. "The people that are supervising it (a project) are more dependent on their ability to judge people than they are dependent on their ability to judge the work that is going on," Noyce said in 1965. He avoided Shockley's mistakes. He established a very casual and open working environment, where his brilliant young employees enjoyed working and worked with responsibility.

During one of the last interviews, he was asked what he would do if he becomes the 'emperor' of the United States. He answered that he would, amongst other things, "make sure we are preparing our next generation to flourish in a high-tech age. And that means education of the lowest and the poorest, as well as at the graduate school level." ●

The author has written six science books published by Pustak Mahal, New Delhi and an engineering book by Industrial Press, New York. Radio Talker and RCFA specialist, he is presently the head of technical training with Coromandel International Ltd for their group of companies.

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